

Programming in Python for Friends and Relations, Part 11



Secure Communication

Here's a simple application to help us stop making silly mistakes while communicating over e-mail.

The faint smile on your face turns into an expression of panic the moment you check your e-mail. "Oh, my God! The CEO has just forwarded an e-mail to Kiran, a union leader, instead of Kiran, the CFO." But you manage to save the day. The recipient is on leave, and you delete the mail from his inward queue.

Your HR chief asks you to check your e-mail. When you do so, you can't help but smile. The CEO has just declared a holiday for his birthday! But the smile doesn't last long. The HR chief asks you to find out who sent that e-mail, as it surely wasn't the CEO. You are reminded of the line routinely printed by banks on their statements: "This is a computer-generated statement and does not require a signature!"

Increasingly, your financial dealings are online.

The statements are being sent by e-mail. To minimise the chances of the wrong person viewing confidential information, the statements are password protected. However, the passwords aren't very strong. They protect against casual snooping, which is fine for most of us. But you do need to put some effort into figuring out the password for each statement. It's not easy for you and your colleagues to do so.

There has to be a better way to manage all these scenarios.

Public key infrastructure

Public key-based algorithms have been around for about as long as I have been in the software field. Ubuntu owes its existence to the money made from the sale of Thawte, which issues digital certificates.